

ECE 230

Practice Final

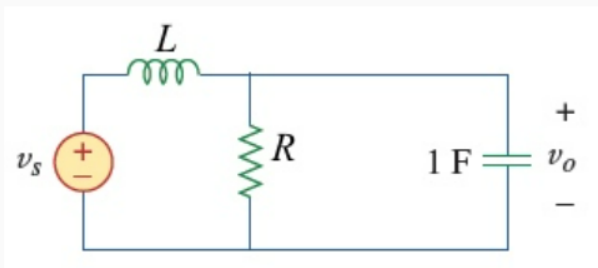
Exam Length: 2 Hours

There will be a total of 10 different problems. What follows is a good representation of what could be on the final. I tried to keep the time/level difficulty comparable to what you will find on the final. I recommend you time yourself, complete the exam in 2 hrs., and THEN check your answers.

Problem 1:

In the circuit given below, $R = 0.25 \Omega$ and $L = 10 \text{ H}$.

NOTE: This is a multi-part question. Once an answer is submitted, you will be unable to return to this part.



Determine the transfer function V_o/V_s , and what kind of filter this is.

Problem 2:

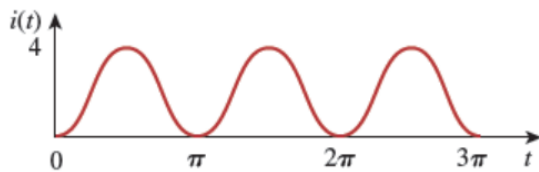
An ideal low-pass filter (LPF) has a critical frequency of 200 rad/s . The passband gain is 1. If the input to the filter is

$$v_i = 2 + 3\cos(150t) + 4\sin(250t+45^\circ) + 5\cos(350t-45^\circ) \text{ V}$$

which terms would appear in the expression for the output voltage?

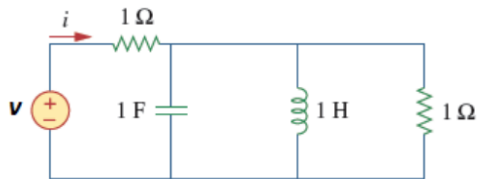
Problem 3:

Find the rms value of the offset sine wave shown in the given figure.



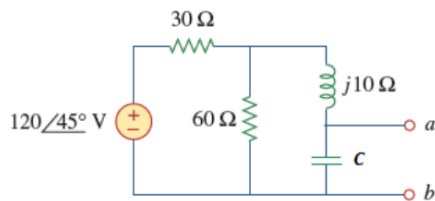
Problem 4:

Determine the current i in the circuit given below, where $v(t) = 9 \cos(10t)$ V. Please report your answer so the magnitude is positive and all angles are in the range of negative 180 degrees to positive 180 degrees.



Problem 5:

Find the Thevenin and Norton equivalent circuit at terminals a - b for the circuit shown below, where $Z_C = -j5\ \Omega$. Please report your answer so the magnitude is positive and all angles are in the range of negative 180 degrees to positive 180 degrees.



Problem 6:

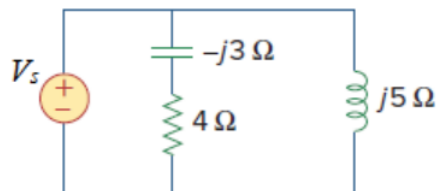
An ac motor with impedance $Z_L = (2.4 + j2.60)\ \Omega$ is supplied by an RMS value of 220-V, 60-Hz source.

Find the power factor, the average power, and the reactive power.

Determine the capacitor required to be connected in parallel with the motor so that the power factor is corrected to unity.

Problem 7:

For the network in the figure, find the complex power absorbed by each element. Take $V_s = 9\angle -20^\circ\text{ V rms}$.



The complex power supplied by the source is

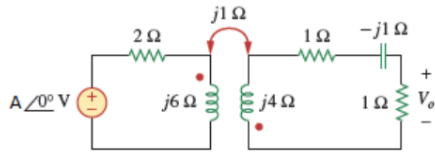
The complex power absorbed by the capacitor is

The power absorbed by the resistor is

The complex power absorbed by the inductor is

Problem 8:

For the given circuit, find V_o if $A = 24$. Please report your answer so the magnitude is positive and all angles are in the range of negative 180 degrees to positive 180 degrees.



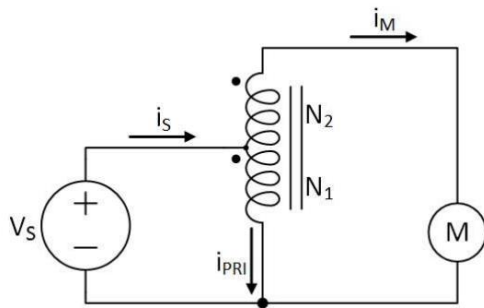
Problem 9:

A stereo amplifier circuit with an output impedance of $7.2 \text{ k}\Omega$ is to be matched to a speaker with an input impedance of 8Ω by a transformer whose primary side has 3240 turns. Calculate the number of turns required on the secondary side.

Problem 10:

For the questions on this page, assume that that an induction motor is rated to deliver **10kW** (approximately **13.3HP**) when operated at **240V_{RMS}** 60Hz. The motor has a lagging power factor of **0.8**.

The building electrical service only supplies **208V_{RMS}**, so the motor can not be connected directly to the electric service.



What is the required turns ratio (N_2/N_1) required?